

OUTPUT:

SGD

0 14.008757808380796

10 0.12492532034396209

20 0.01151353986194053

30 0.010579587812703024

40 0.010571785931001111

RMSprop

0 14.008757808380796

10 12.229517844972454

20 11.159852980829587

30 10.223506647671746

40 9.3548933128694

Adam

10 13.059127647433067

20 12.050945547657971

30 11.096818044286838

40 10.19908649391244

50 9.358007974120348

OUTPUT:

Training with Sigmoid

Epoch 20, Loss = 0.9879

Epoch 40, Loss = 0.7602

Epoch 60, Loss = 0.6310

Epoch 80, Loss = 0.5401

Epoch 100, Loss = 0.4727

Training with Tanh

Epoch 20, Loss = 0.4644

Epoch 40, Loss = 0.4177

Epoch 60, Loss = 0.4172

Epoch 80, Loss = 0.4172

Epoch 100, Loss = 0.4172

Training with ReLU

Epoch 20, Loss = 0.1989

Epoch 40, Loss = 0.0316

Epoch 60, Loss = 0.0050

Epoch 80, Loss = 0.0008

Epoch 100, Loss = 0.0001

OUTPUT:

Feature matrix (x_train): (1500, 8, 8)

Target matrix (y_train): (1500,)

Feature matrix (x_test): (297, 8, 8)

Target matrix (y_test): (297,)

